

Physical-Model-Based Optical Line System Transmission Optimization Integrating Probing Transceivers

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Abstract An automated methodology for optical line systems integrating probing transceivers to initialize a system-level digital twin is described. Physical model parameters are inferred from scratch by combining end-to-end spectrum measurements, pre-FEC BER and optical amplifier telemetry, enabling near-optimal configuration, model alignment, and precise QoT tuning. ©2026 The Author(s)

Introduction

The rapid proliferation of artificial intelligence applications is driving an unprecedented demand for connectivity, increasing traffic volumes and imposing stringent requirements on network scalability [1]. To cope with this growth, network operators must deploy infrastructure rapidly while overcoming the limitations of vertically integrated solutions, which introduce vendor lock-in and reduce flexibility [2]. As a result, fiber-optic transport networks are evolving towards open and disaggregated architectures [3]. In this context, operators are transitioning towards partially disaggregated optical networks, while more advanced paradigms such as multi-domain or fully disaggregated networks are still emerging. These scenarios require a centralized control paradigm and a common quality-of-transmission (QoT) metric, such as the generalized signal-to-noise ratio (GSNR), to enable consistent end-to-end management [4]. This requires models capable of accurately evaluating transmission performance across heterogeneous domains, which is challenging due to limited system knowledge. The Telecom Infra Project has introduced GNPpy for QoT estimation based on physical-layer models, but its applicability depends on accurate model alignment with the deployed system [5]. At the same time, data-driven and physics-informed approaches are emerging, but still rely on a robust physical baseline [6].

In this work, we propose an automated commissioning methodology for optical line systems (OLSs) integrating probing transceivers, aimed at initializing a system-level digital twin and enabling system optimization by maximizing GSNR while ensuring spectral flatness [7]. The approach enables the inference of physical model parameters from scratch by combining end-to-end spectral

measurements, bit error rate (BER), and optical amplifier telemetry, allowing the construction of a consistent representation of the deployed system without requiring detailed prior knowledge of its physical characteristics. Finally, it establishes a robust physical-model baseline for QoT estimation, supporting accurate model alignment and enabling the integration of AI-based optimization frameworks for the optical network digital twin.

Experimental Setup

The considered transmission scenario is an OLS connecting two adjacent nodes within a mesh optical network (Fig. 1-a). The nodes are equipped with reconfigurable optical add-drop multiplexers, including a booster (BST) at the transmitter side and a pre-amplifier at the receiver side, interfacing the OLS with the surrounding network. The OLS consists of a cascade of standard single-mode fiber spans, each followed by an inline amplifier (ILA) implemented as a dual-stage erbium-doped fiber amplifier (EDFA). At the transmitter side, the spectral load is generated by an amplified spontaneous emission (ASE) source, spectrally shaped and multiplexed into the OLS, while at the receiver side the propagated spectrum is demultiplexed and observed by an optical spectrum analyzer (OSA). A probing transceiver (TRX) operating at the central frequency is added at the transmitter side and dropped at the receiver side. Along the line, each EDFA operates under automatic gain control, allowing the configuration of gain and spectral tilt, and integrates total power monitors at its input and output terminals. The experimental setup implements 5 standard single-mode fiber spans for a total length of approximately 400 km. The ASE source emulates a fully loaded C-band WDM spectrum with 40 channels on a 100 GHz grid and 50 GHz bandwidth. The TRX is centered

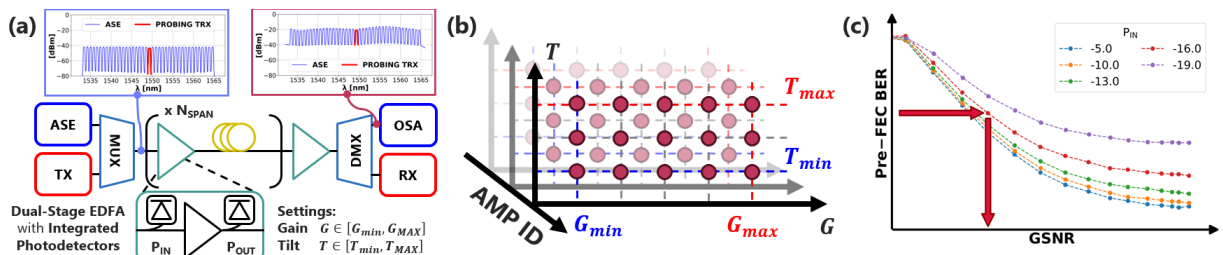


Fig. 1: (a) OLS experimental laboratory setup configuration; (b) dataset acquisition procedure for model calibration; (c) conversion of the pre-FEC BER measured by the transceiver into GSNR using the back-to-back characterization curve.

at 193.5 THz and operates at 63.1 GBd using a DP-16QAM modulation format (400 Gb/s). The EDFAs operate over a gain range of 10–20 dB and a tilt range of 0 to -2 dB, with a maximum output power of 20 dBm.

Dataset Acquisition

The dataset acquisition procedure is performed by probing the OLS under different operating conditions [8], starting with a reference transparency configuration in which each amplifier compensates for the loss of the preceding span. A *first dataset* is obtained using a swept single-channel probing load to capture the frequency-dependent attenuation on a span-by-span basis and measuring the EDFA total power monitors, allowing the contribution of inter-channel stimulated Raman scattering (SRS) to be considered negligible. At this stage, the probing TRX also enables the measurement of the accumulated chromatic dispersion as a function of frequency. A *second dataset* is collected using a fully loaded WDM spectrum with the TRX wavelength placed at the center, where both signal and noise spectra are observed at the end of the line. Starting from the reference configuration, each amplifier is sequentially perturbed by sweeping the gain and tilt settings within their operating ranges according to a predefined sampling grid (Fig. 1-b). For each configuration, the corresponding telemetry is recorded, resulting in a dataset that captures the system sensitivity to the applied parameter variations. A single set of telemetry measurements consists of: i) total input and output power monitored at each amplifier; ii) the optical spectrum measured at the receiver; iii) the received power and pre-FEC BER of the probing TRX. The pre-FEC BER is mapped to an equivalent GSNR using the corresponding back-to-back characterization curve [9], obtained for different received power levels, P_{IN} (Fig. 1-c) [10].

Model Calibration

The OLS digital model combines a fiber propagation model with a parametric amplifier model,

enabling end-to-end simulation of signal and noise impairment evolution along the line. The fiber propagation is modeled using the open-source GNPpy framework [11], which implements a Gaussian noise model for nonlinear interference (NLI) and accounts for SRS effects through dedicated nonlinear solvers, enabling frequency-dependent evaluation of signal power and noise accumulation [12, 13]. Each EDFA is described by a parametric model of the gain and noise figure spectra as functions of the operating settings, namely gain and spectral tilt. Extending the modeling approach presented in [14], the amplifier gain profile is modeled using a Taylor-like polynomial expansion, least-squares fitted, including up to second-order terms in gain and first-order terms in tilt. Similarly, the noise figure is modeled as a function of gain and tilt using a first-order expansion [15].

The calibration procedure aims at identifying the physical parameters of the OLS model from the acquired dataset through a sequential parameter extraction strategy. First, the frequency-dependent attenuation and chromatic dispersion are estimated using the first dataset. The attenuation profile is derived on a span-by-span basis from the measured power levels, while the dispersion is obtained by selecting the most suitable fiber type from a predefined database through a brute-force matching procedure [16]. The fiber lengths, and intra-span lumped losses intensities and positions are assumed to be known from OTDR reports. Then, the attenuation is then decomposed into loss coefficient function and connector losses at the span boundaries using the second dataset. Two key aspects enable the independent estimation of the fiber parameters. First, the availability of both GSNR (from the TRX) and SNR_{ASE} (from the OSA) allows the separation of NLI from ASE, thus decoupling fiber-related nonlinear effects from amplifier noise contributions. Second, the joint contribution of SRS and NLI provides a unique balance of nonlinear effects, enabling con-

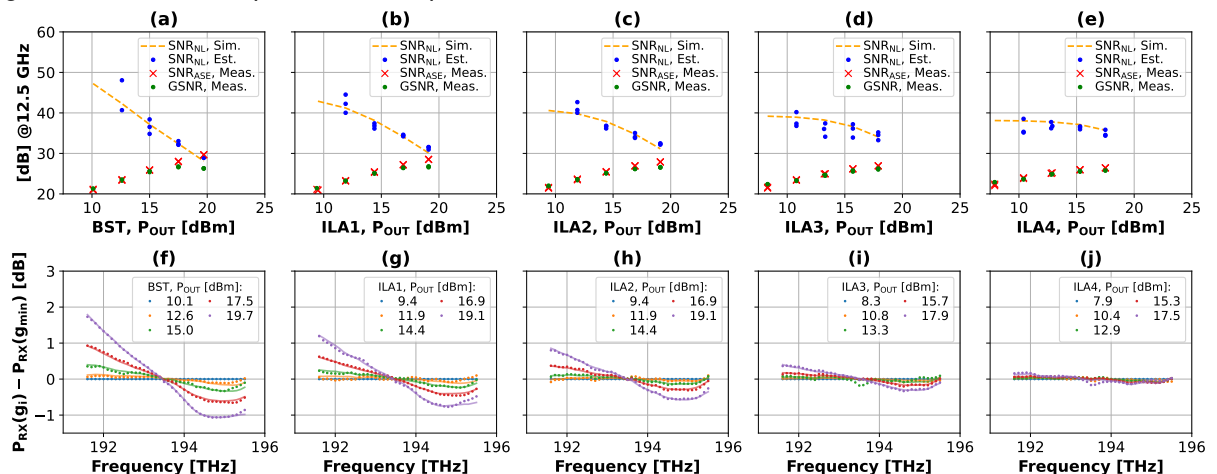


Fig. 2: Model calibration. Observing the OLS termination and changing a single OA gain, from the BST to the last ILA, (a–e) measured and simulated nonlinear impairment, (f–j) measured and simulated SRS-induced residual spectral ripple.

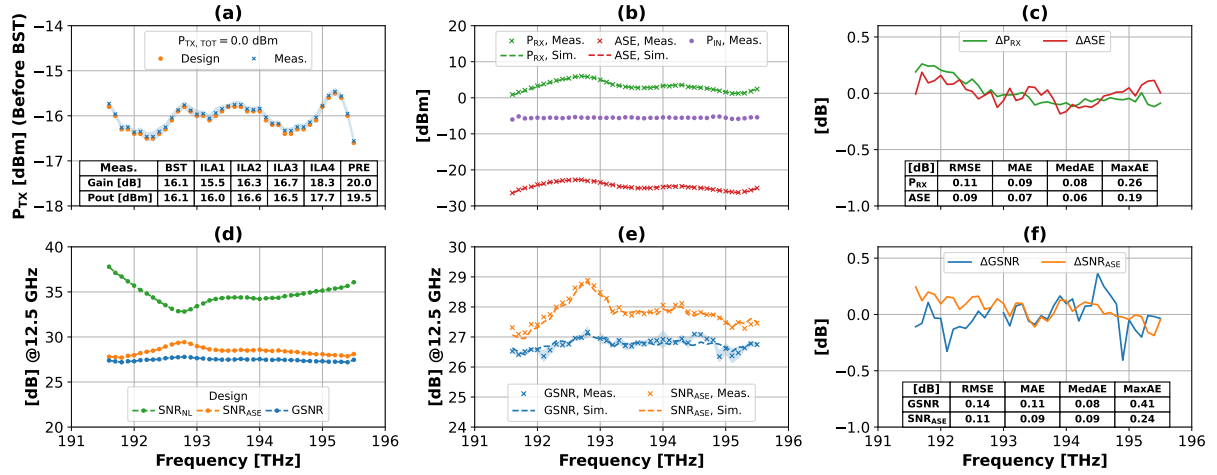


Fig. 3: OLS optimization. (a) Launch power profile; (b) received signal and ASE power, and measured transceiver input power after equalization; (c) error between measured and simulated received signal and ASE power; (d) QoT metrics computed at design phase; (e) measured and simulated SNR_{ASE} and GSNR spectra; (f) error between measured and simulated SNR_{ASE} and GSNR.

sistent identification of the fiber parameters for each span. Once the fiber parameters are determined, the amplifier gain model is calibrated. The partial derivatives of the gain profile with respect to gain and tilt are extracted, together with the intrinsic gain ripple, assumed to be identical for all amplifiers. Finally, the noise figure parameters are estimated, including both the average noise figure dependence on the operating conditions and its partial derivatives with respect to gain and tilt. The parameter estimation is performed using the L-BFGS-B method [17], minimizing the absolute error between measured and simulated quantities, including the received signal and ASE power profiles, as well as the GSNR of the central channel. Results of the calibration with respect to the acquired dataset are reported in Fig. 2, using a sampling grid with 5 and 3 values for gain and spectral tilt, respectively. By varying the gain of each amplifier, from the BST to the last ILA, the results show the agreement between measured (markers) and simulated (lines) nonlinear impairment, Fig. 2-(a–e), as well as the SRS-induced residual spectral ripple, Fig. 2-(f–j).

OLS Optimization

Once the digital twin is initialized, the optimization of the OLS operating point is performed in two sequential steps. Firstly, the operating point of the amplifiers is determined under a flat launch power spectrum assumption. The gain and tilt settings are then optimized under this constraint, such that the BST can operate at its maximum output power without entering saturation, while maximizing the overall GSNR and minimizing its standard deviation across the spectrum [18]. In the second step, the amplifier configuration is fixed, and the optimal spectral equalization is determined. The channel power distribution is adjusted to achieve a spectrally flat GSNR profile, compensating for residual frequency-dependent impairments. This

two-step procedure enables the identification of a near-optimal operating condition for the OLS, balancing performance maximization and spectral uniformity. The optimized configuration is then translated into the corresponding device settings and applied to the physical system, enabling validation through experimental measurements. Results of the optimization procedure are reported in Fig. 3. The validation is carried out by sweeping the TRX across all 40 WDM channels, measuring the received power and pre-FEC BER at each frequency to estimate the GSNR profile over the entire C-band. For each channel, 10 repeated measurements are performed. The comparison between measured and simulated quantities, including signal and ASE power as well as SNR_{ASE} and GSNR spectra, yields a mean absolute error (MAE) of approximately 0.1 dB across all metrics.

Conclusion

In this work, an automated commissioning methodology for optical line systems has been presented, enabling the joint calibration and optimization of the OLS operating point from system-level measurements. Experimental results demonstrate the effectiveness of the proposed methodology in accurately aligning the model with the physical system and achieving consistent performance optimization, even under limited system knowledge conditions that challenge consistent QoT estimation. The proposed approach minimizes design margins and uncertainty contributions related to the static QoT, with physical parameters, including nonlinear effects, identified assuming operation close to zero-margin conditions [19]. This shifts the margin definition towards system and unlocated margins, which capture dynamic contributions evolving over the network lifetime. Data-driven and physics-informed approaches can be leveraged to minimize these uncertainties, paving the way towards scalable control frameworks and optical network digital twins.

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